

Silverline House — 25-Storey Residential HRB

External Wall Fire Performance Assessment

Document Reference: 2.2.6
Revision: Rev A
Date: March 2026
Project Reference: SLH-2026

1 Introduction

This Fire Strategy Report has been prepared to accompany the Gateway 2 application for Building Control Approval for Silverline House, a 25-storey residential higher- risk building. The strategy has been developed in accordance with Approved Document B (Fire Safety) Volumes 1 and 2, BS 9999:2017, and BS 9991:2015. The report identifies and justifies all elements of the fire safety design including means of escape, passive fire protection, active fire suppression, fire detection, and fire service access. The strategy is supported by evacuation simulations carried out in accordance with PD 7974-6 and confirmatory computational fluid dynamics (CFD) modelling of the smoke extraction system. This assessment evaluates the fire performance of the external wall system in accordance with Approved Document B clause 12.7 and the MHCLG Advice Note 14.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

2 System Description and Test Evidence

The rainscreen cladding system has been selected and specified to comply with the requirements of Approved Document B (external fire spread), Approved Document C (resistance to weather), and Approved Document L (energy efficiency). The system comprises fibre cement panels in an open-joint configuration, a ventilated cavity, mineral wool insulation boards, an air and vapour control layer (AVCL), and the primary wall construction. The system has been tested in accordance with BS 8414-1 and assessed against BR 135 Fire Performance of External Thermal Insulation for Walls of Multistorey Buildings. Third-party certification is provided by the BBA. The system has been assessed against BR 135 following testing to BS 8414-1. The test report reference is BRE-2025-SLH-001. The system achieves a pass result with no observations of unlimited fire spread.

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3 Compliance Statement

The external wall system for Silverline House complies with the requirements of Approved Document B and the MHCLG External Wall System advice. The system uses only materials classified A2-s1,d0 or better in the outer leaf. The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

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